

Program : Diploma in Printing Technology	
Course Code : 6102C	Course Title: Security Printing
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge on the Purpose and features of security printing.
- To guide the students to identify the security feature of a product.
- To provide the knowledge on the various application of Security printing methods.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Optics		Applied Physics- II	II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain security printing process.	15	Understanding
CO2	Describe Security inks and substrates.	14	Understanding
CO3	Illustrate currency printing, cheque printing, numbering and bar-coding.	15	Understanding
CO4	Summarize Holograms, Credit Cards and passports security features.	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the stages of printing process.		
M1.01	Security printing – Introduction.	3	Understanding
M1.02	Security features.	4	Understanding
M1.03	Basic principles of security design features.	4	Understanding
M1.04	Various printing methods used for security printing.	4	Understanding
Contents: Security printing - Definition. Goals of security printing. Security features - water mark, security thread, latent image, micro lettering, Intaglio Printing, identification mark, see through register, hologram etc. Basic principles of security design features - Pantograph screens, Void pantograph Screen, ODT - optical deterrent technology, Guilloches, Warning bands, Code safe, High resolution graphics and Padlock icon. Various printing methods used for security printing.			
CO2	Describe Security inks and substrates.		
M 2.01	Features of security inks.	4	Understanding
M 2.02	Features of security papers.	3	Understanding
M 2.03	Types and usages of security threads.	4	Understanding
M 2.04	Watermark.	3	Understanding
	Series Test – I	1	
Content: Features of security inks - Trademark colors, Color changing ink, Magnetic ink, Copy protection ink, Erasable ink, Fugitive ink, Pen reactive ink, Heat reactive ink, Coin reactive inks, Migrating ink, Bleeding inks, Florescent ink, Metallic ink and UV ink etc. Features of security papers - Safety paper, Chemical reactive paper, Special papers, Water mark paper and Copy evident paper etc. Types and usages of security threads - Metalized thread, Windowed thread, Holographic windowed thread, Micro text, Clear text and Thermo			

text. Basic principles of watermark, Classification of watermark - Line drawing watermark, shaded watermark and Digital watermark.

CO3	Illustrate currency printing, cheque printing, numbering and bar-coding.		
M3.01	Bank Note security features.	5	Understanding
M3.02	Cheque printing.	5	Understanding
M3.04	Barcode.	5	Understanding

Contents:

Creation & Graphics, Bank Note security features, Applications of Computers in Designing. Currency Signatures & numbering, Paper Specifications. Watermark & Other Protective devices, Digital Watermark on Currency, Circulation & Bank Maintenance. Special issues counterfeiting. Cheque Printing, Numbering And Bar Coding. Introduction, Pre-Encoding and Post Encoding, Printing Tolerances, Testing equipment, E13B Characters, RBI Specifications, Principles of Cheque Design. Numbering with MICR Ink on Rotary presses, Trouble Shooting, Modulus Systems, Weighted & Unweighted. Barcode: Introduction, types- Linear and matrix, barcode structure and characters, Principles of Bar Coding, Types of Coding- EAN 13 Code, Code 39 ACA etc., Typical Bar Code Machines & Print wheels.

CO4	Summarize Holograms, Credit Cards and passports security features.		
M4.01	Hologram -Introduction & Manufacturing	5	Understanding
M4.02	Types of Holograms	5	Understanding
M4.03	Credit card – Security features & Manufacturing	4	Understanding
	Series Test – II	1	

Contents:

Hologram: Introduction, Manufacturing Process, Holographic Recording & Master Origination. Finishing process. Types of Holograms -reflection, transmission and hybrid, Security holograms, Clickograms, stereogram, Anigram etc. Credit card: introduction, security features, Materials used, specifications, Embossing, Magnetic Strip, Recording and Specifications, Manufacturing techniques. Security features in passports.

Text / Reference:

T/R	Book Title/Author
T1	TulikaDas, Chemistry in Printing, 2nd Edition , BarnanaPrakashani, 2011.
R1	P. Kipphan, Handbook of Print Media , Springer, 2002.
R2	Bishwanath Chakravarty, A Hand Book for Printing and Packaging , Galgotia Publications Pvt. Ltd, 1997.
R3	GATF Staff Solving Offset Ink Problems ,GATF, 1998
R4	N.R.Elred& T. Scarlet, What the Printer should know about Ink , GATF, 1995
R5	N.R.Elred& T. Scarlet, Chemistry for the Graphic Arts , GATF, 1992.
R6	R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition , Kluwer Academic publisher.

Online Resources:

Sl.No	Website Link
1	https://www.printwiki.org/Gravure
2	https://www.britannica.com/technology/flexography
3	https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-complete-beginner-c5bd393d9238
4	https://www.britannica.com/technology/offset-printing